DOC334 Week 01

Re-Cap

Learning Outcome of This Session

- LMS Navigation for 2021 MAY [**IMPORTANT**]
- Module overview
- Installing Python 3.x for DOC334 requirements [**IMPORTANT**]
- Flow charts quick re-cap
- Pseudocode quick re-cap
- Python re-cap with a small program
  - Good program practice
  - Marking your programs under DOC334 module

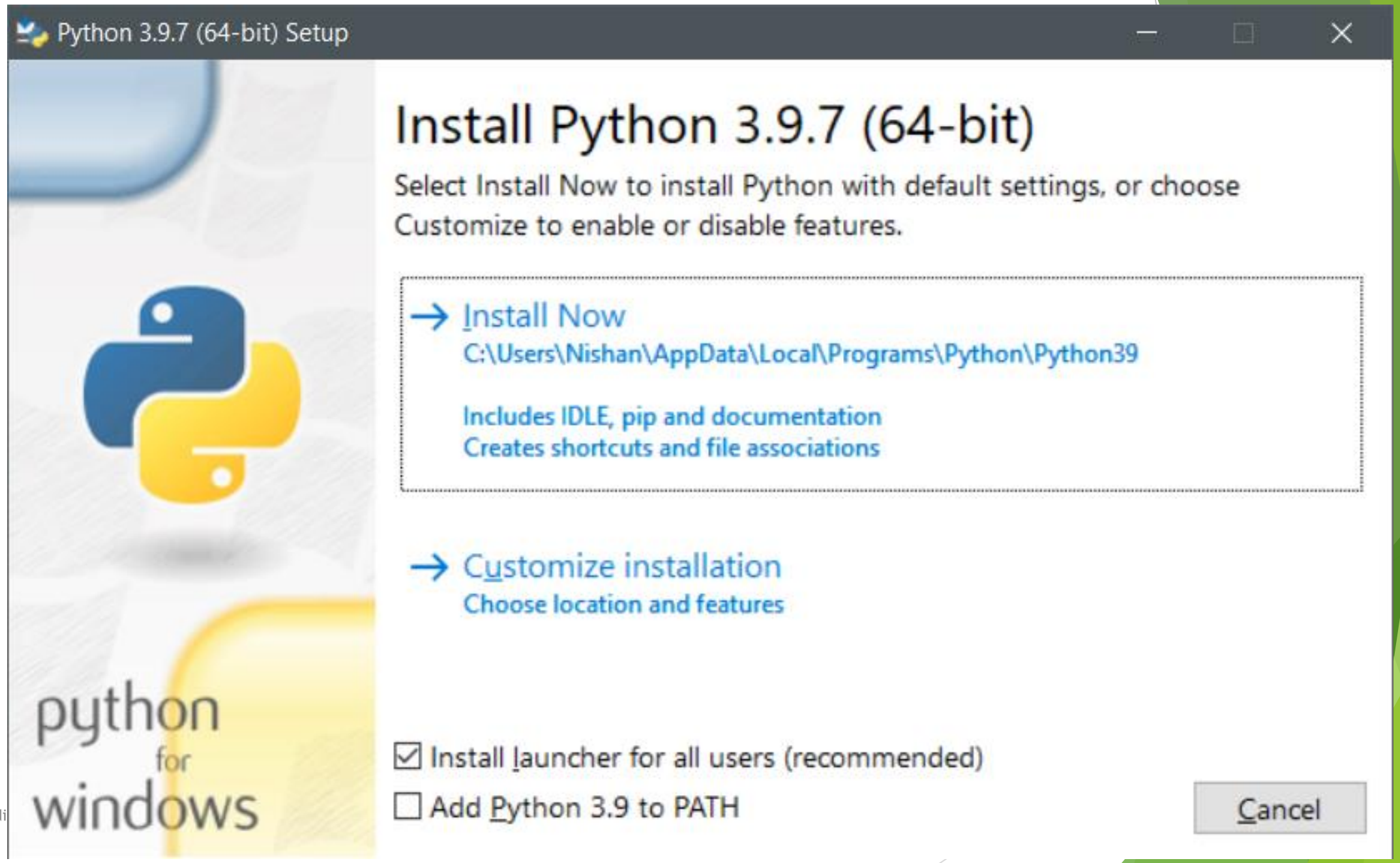
Installing Python 3.x

- These steps are crucial for you to run the DOC334 Python programs in future

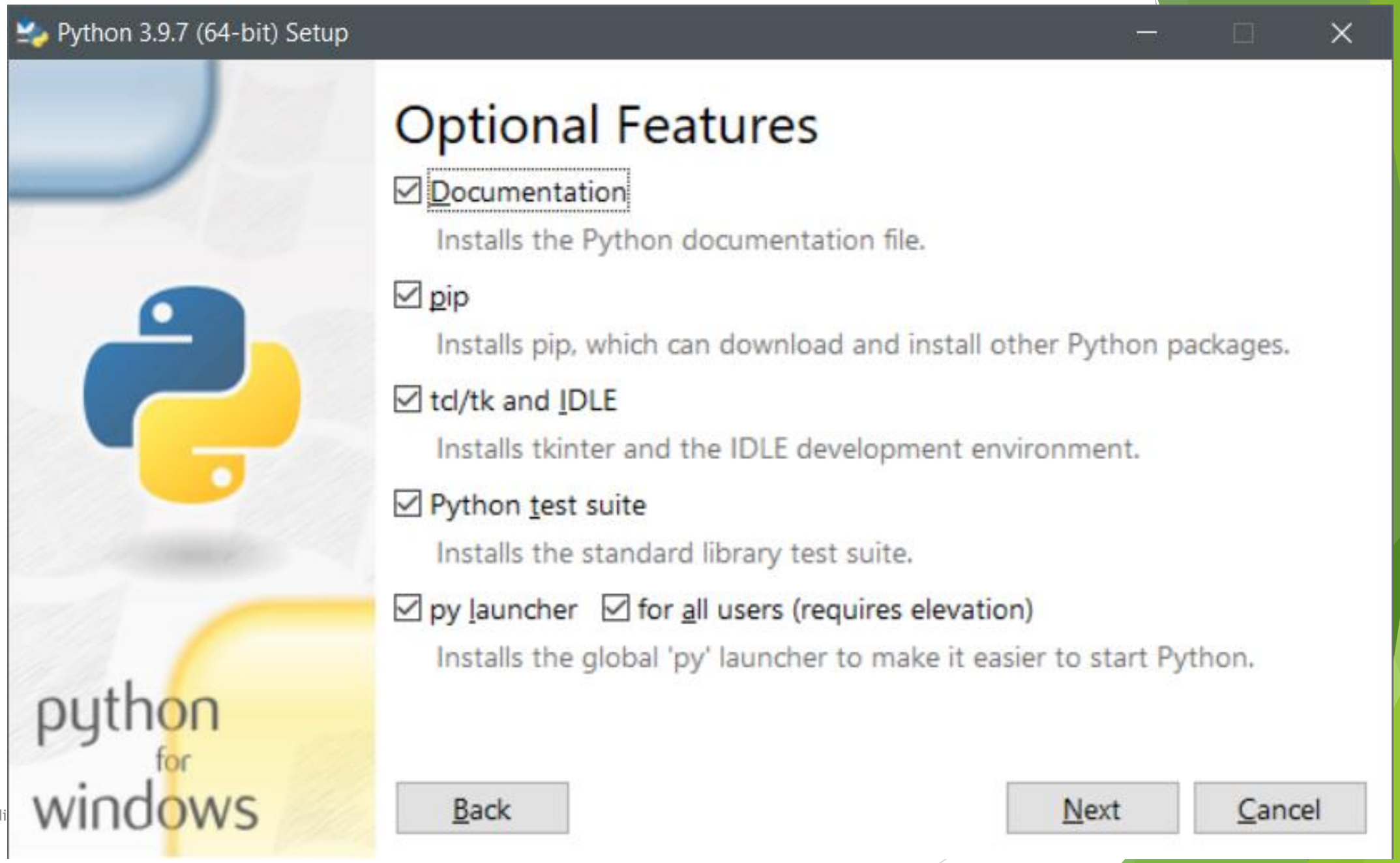
Download from Website

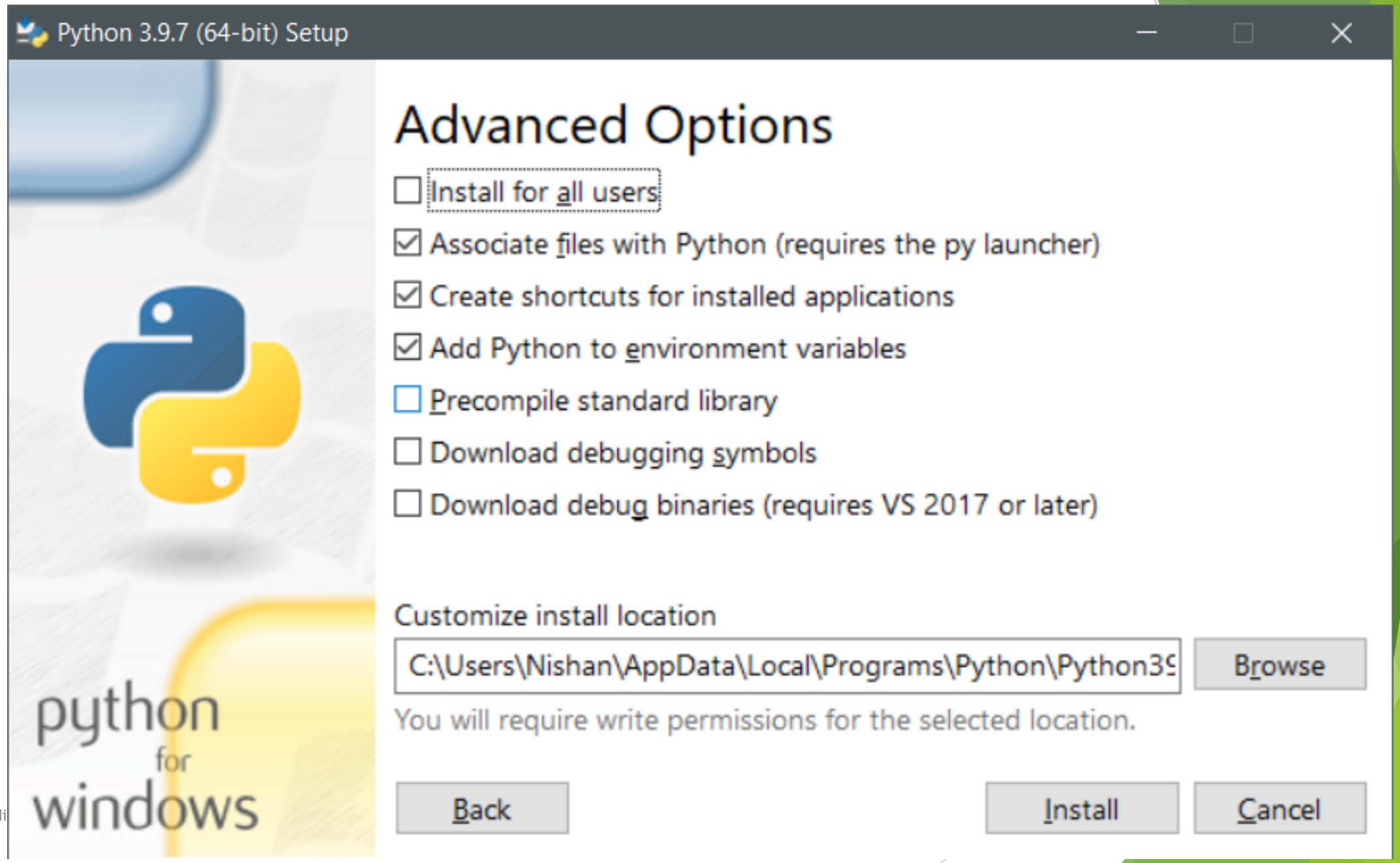
- https://www.python.org/downloads/

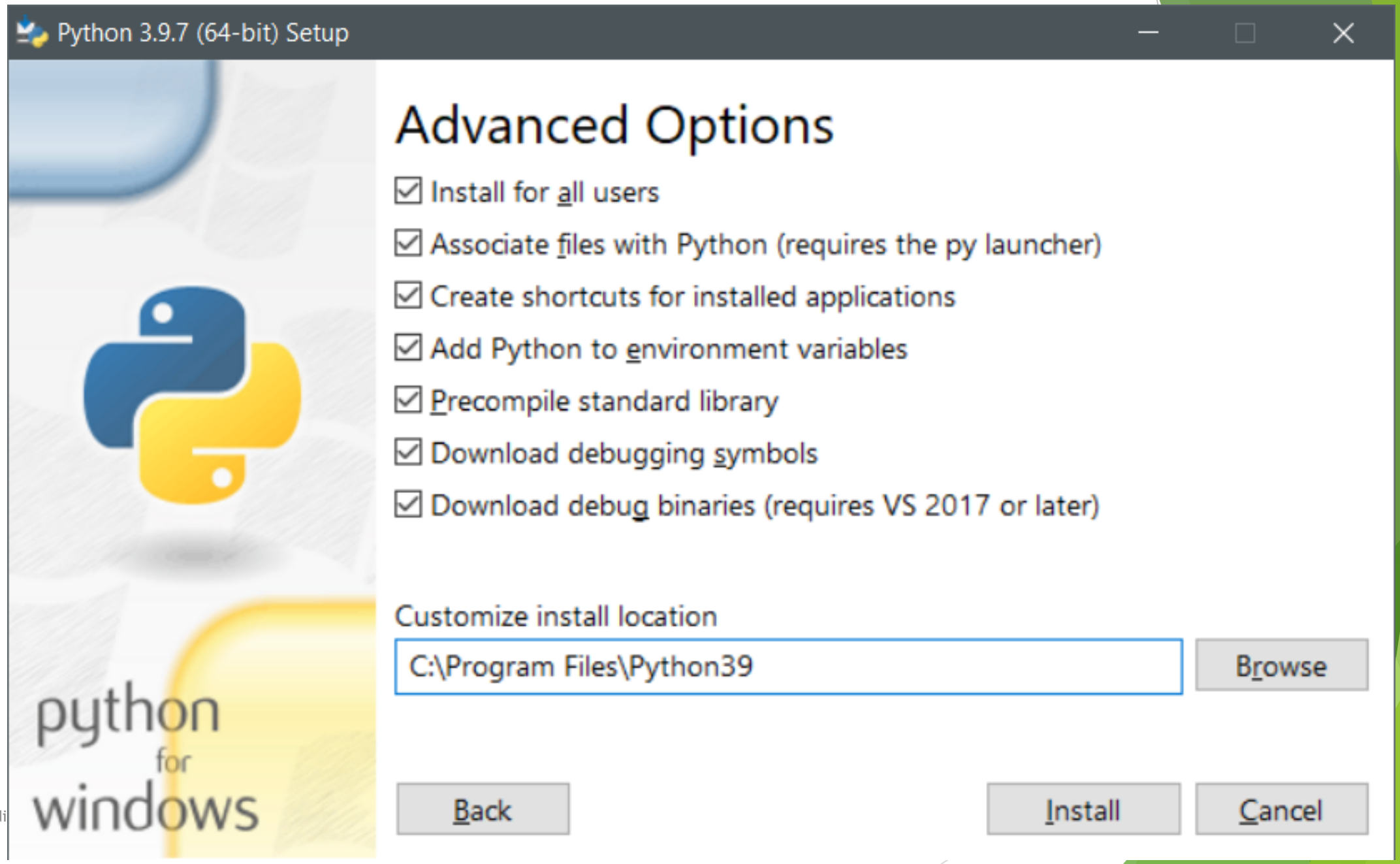


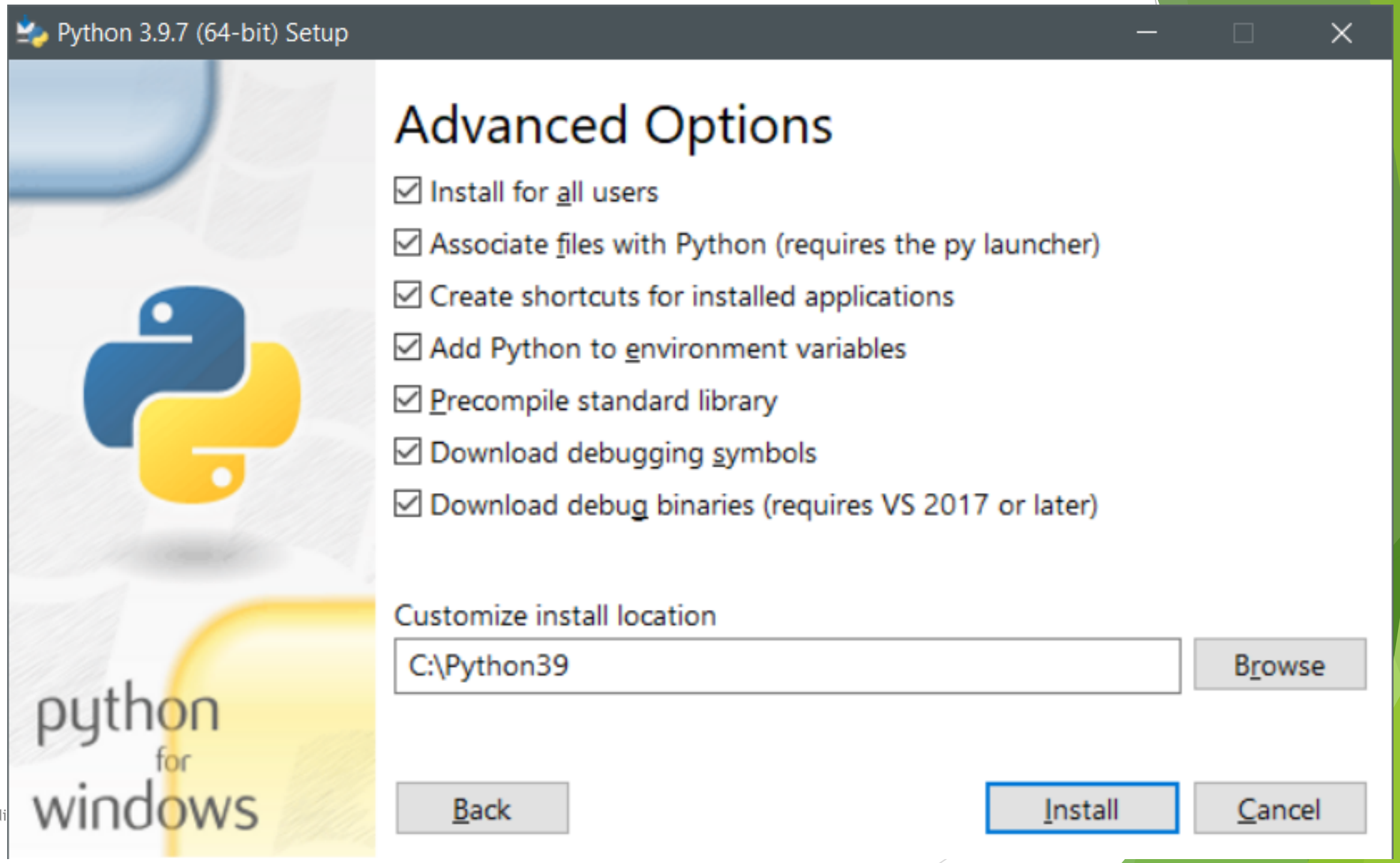

Python 3.9.7 (64-bit) Setup
Install Python 3.9.7 (64-bit)
Select Install Now to install Python with default settings, or choose
Customize to enable or disable features.
Install Now
C:\Users\Nishan\AppData\Local\Programs\Python\Python39
Includes IDLE, pip and documentation
Creates shortcuts and file associations
Customize installation
Choose location and features
python
for
windows
Install launcher for all users (recommended)
Add Python 3.9 to PATH
Cancel

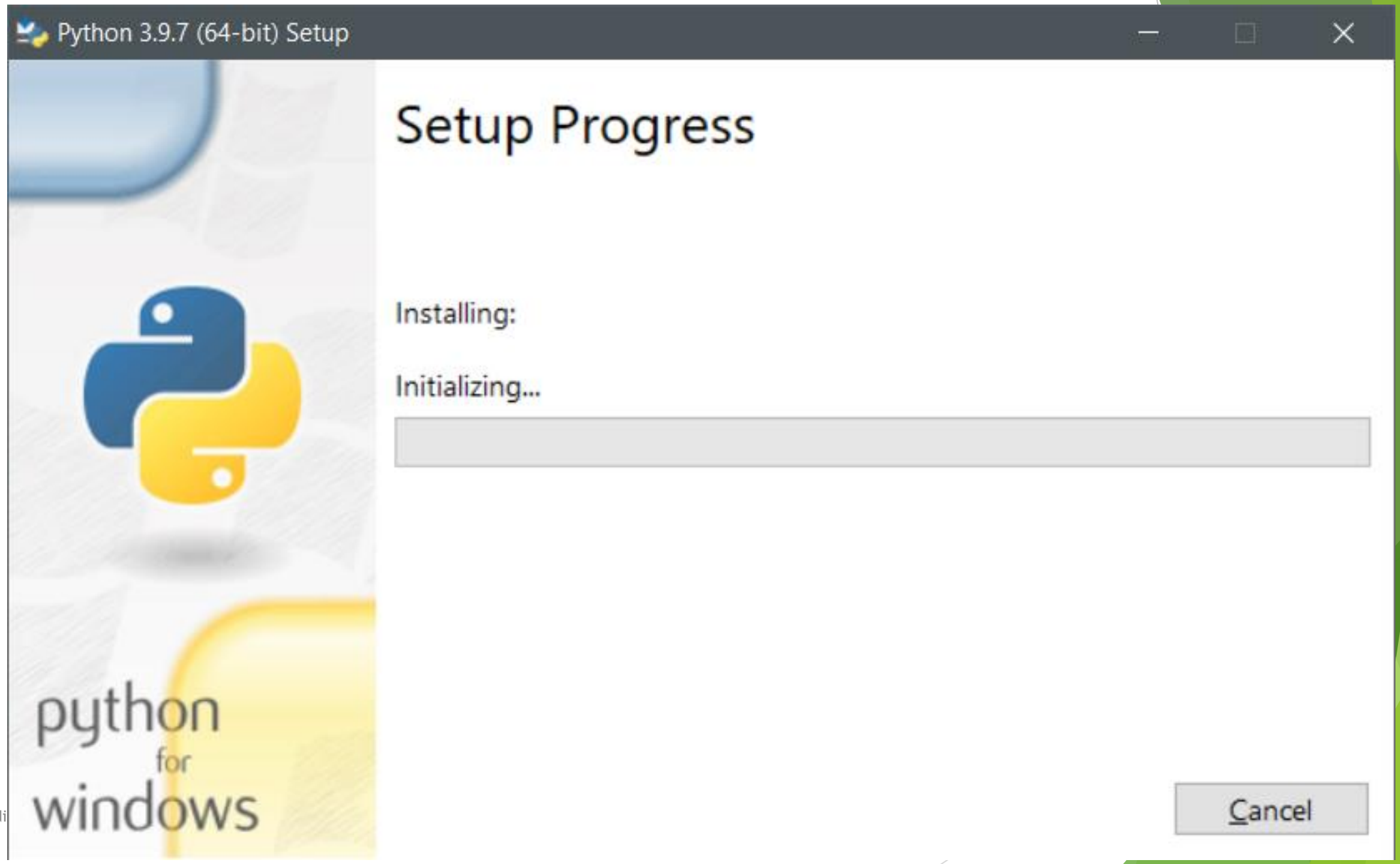

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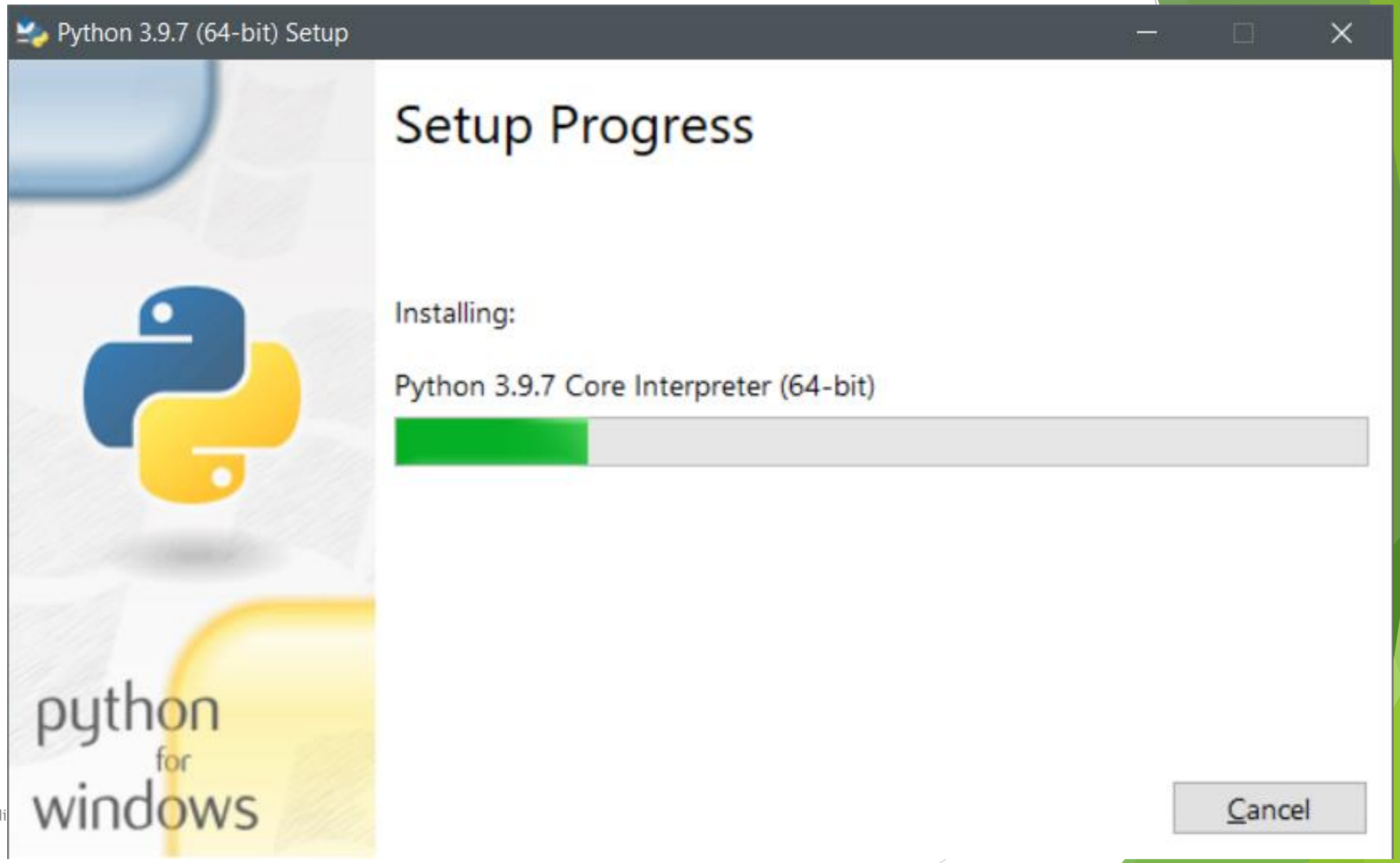

Python 3.9.7 (64-bit) Setup
Optional Features
Documentation
Installs the Python documentation file.
pip
Installs pip, which can download and install other Python packages.
tcl/tk and IDLE
Installs tkinter and the IDLE development environment.
Python test suite
Installs the standard library test suite.
py launcher
for all users (requires elevation)
Installs the global 'py' launcher to make it easier to start Python.
python
for
windows
Back
Next
Cancel

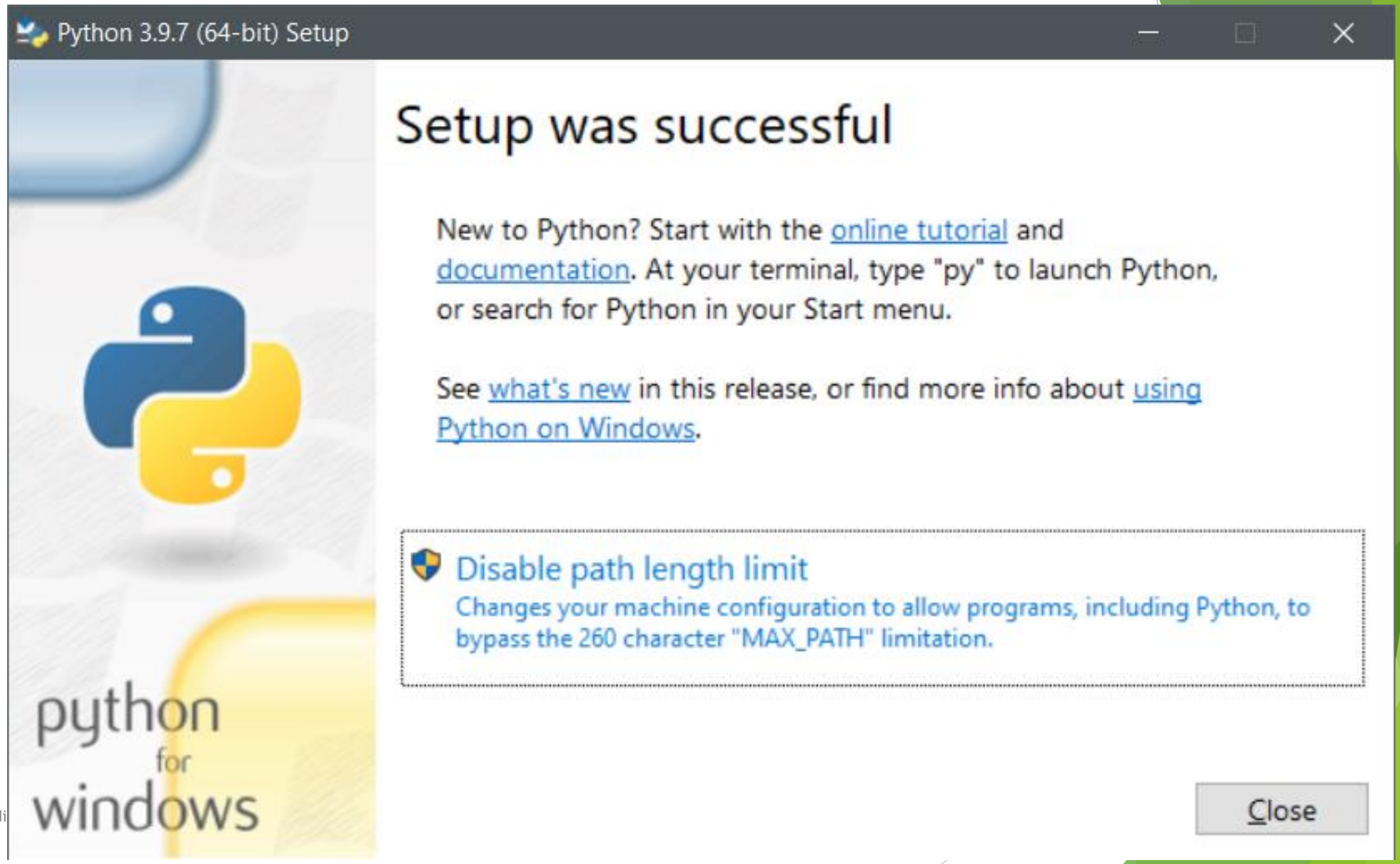

Python 3.9.7 (64-bit) Setup
Advanced Options
Install for all users
Associate files with Python (requires the py launcher)
Create shortcuts for installed applications
Add Python to environment variables
Precompile standard library
Download debugging symbols
Download debug binaries (requires VS 2017 or later)
Customize install location
C:\Users\Nishan\AppData\Local\Programs\Python\Python3
Browse
You will require write permissions for the selected location.
python
for
windows
Back
Install
Cancel

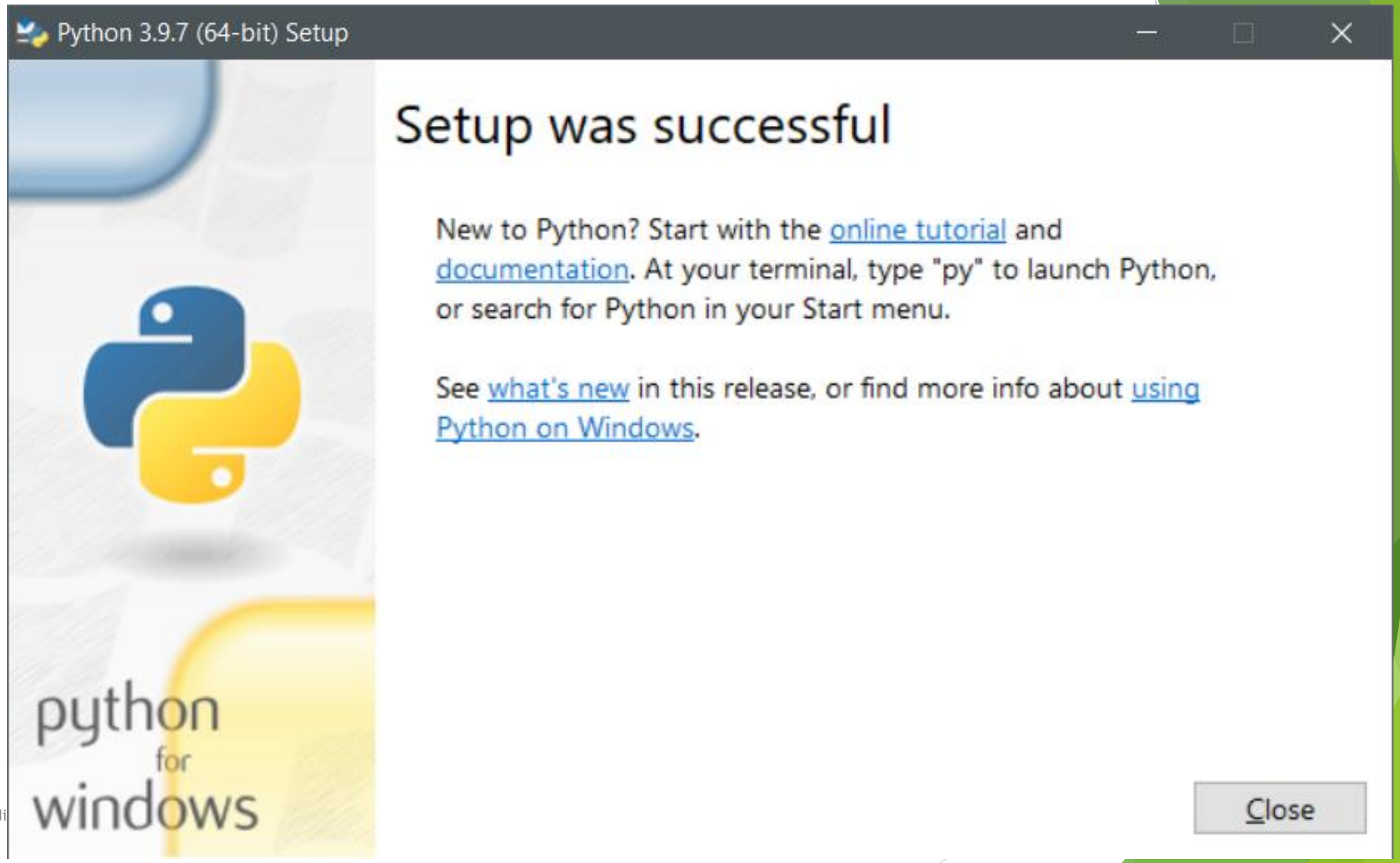

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Customize install location
C:\Program Files\Python39
Browse
python
for
windows
Back
Install
Cancel

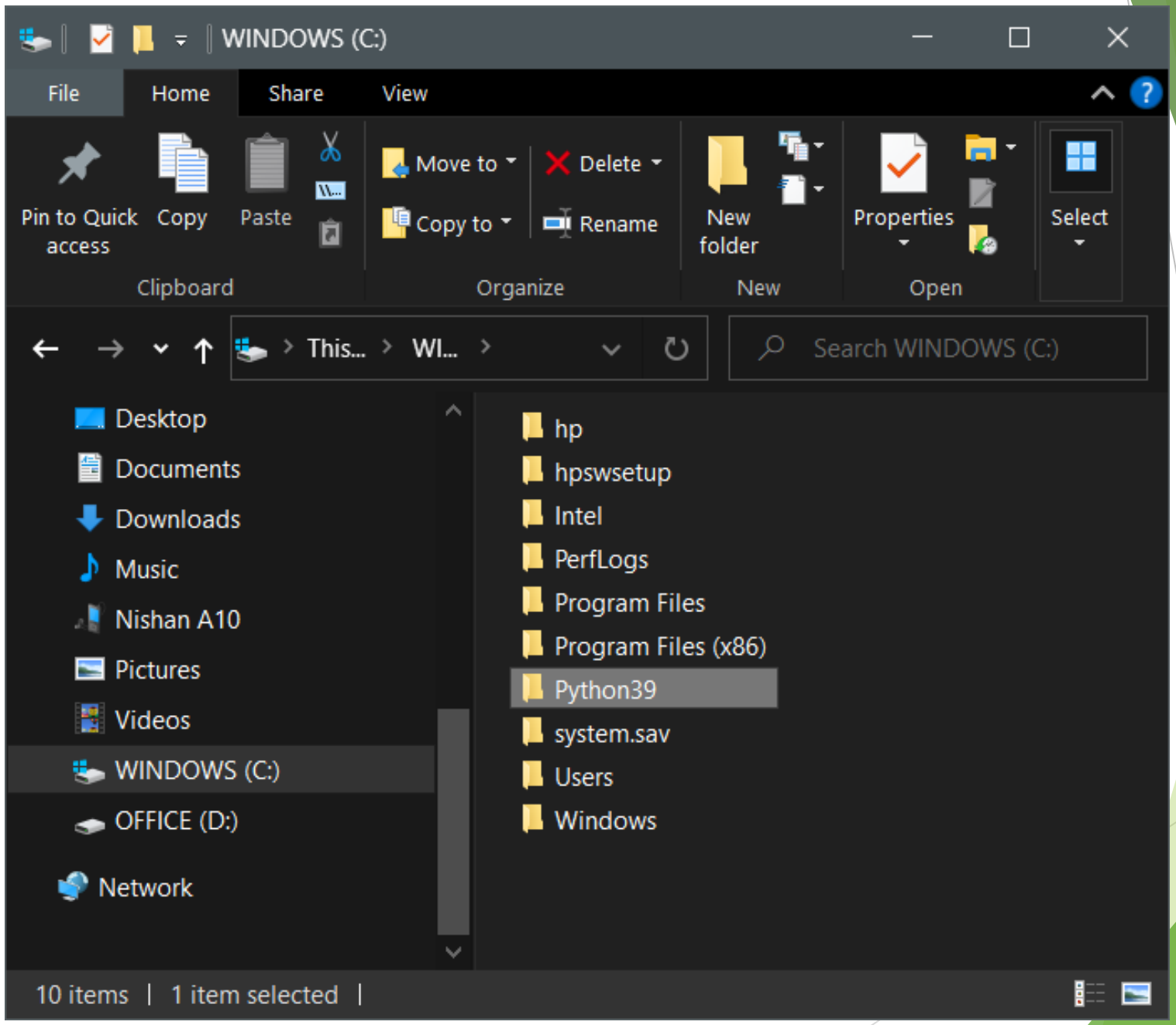

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Customize install location
C:\Python39
Browse
python
for
windows
Back
Install
Cancel


Python 3.9.7 (64-bit) Setup
Setup Progress
Installing:
Initializing...
python
for
windows
Cancel

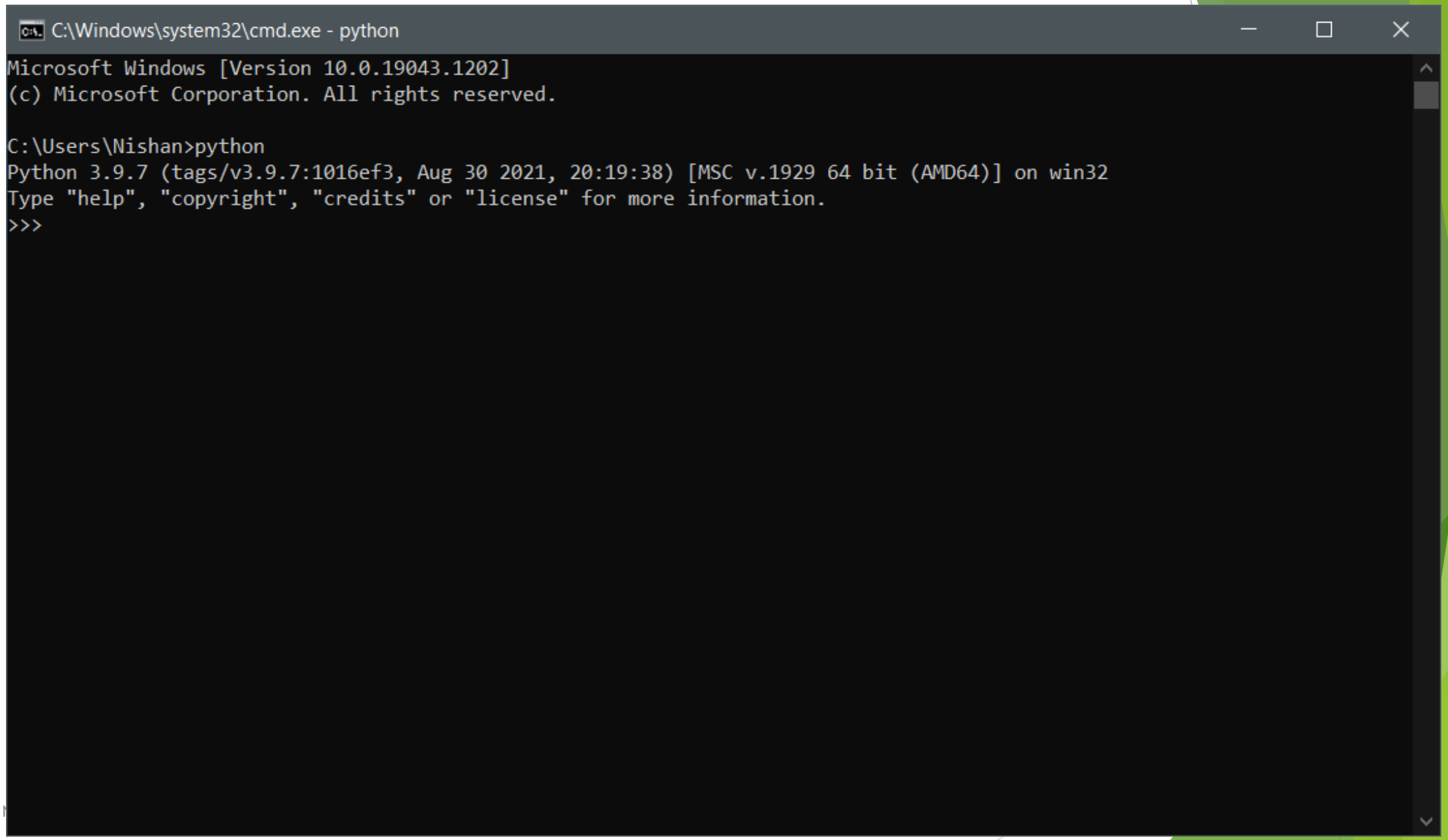

Python 3.9.7 (64-bit) Setup
Setup Progress
Installing:
Python 3.9.7 Core Interpreter (64-bit)
python
for
windows
Cancel


Python 3.9.7 (64-bit) Setup
Setup was successful
New to Python? Start with the online tutorial and documentation. At your terminal, type "py" to launch Python, or search for Python in your Start menu.
See what's new in this release, or find more info about using Python on Windows.
Disable path length limit
Changes your machine configuration to allow programs, including Python, to bypass the 260 character "MAX_PATH" limitation.
python
for
windows
Close

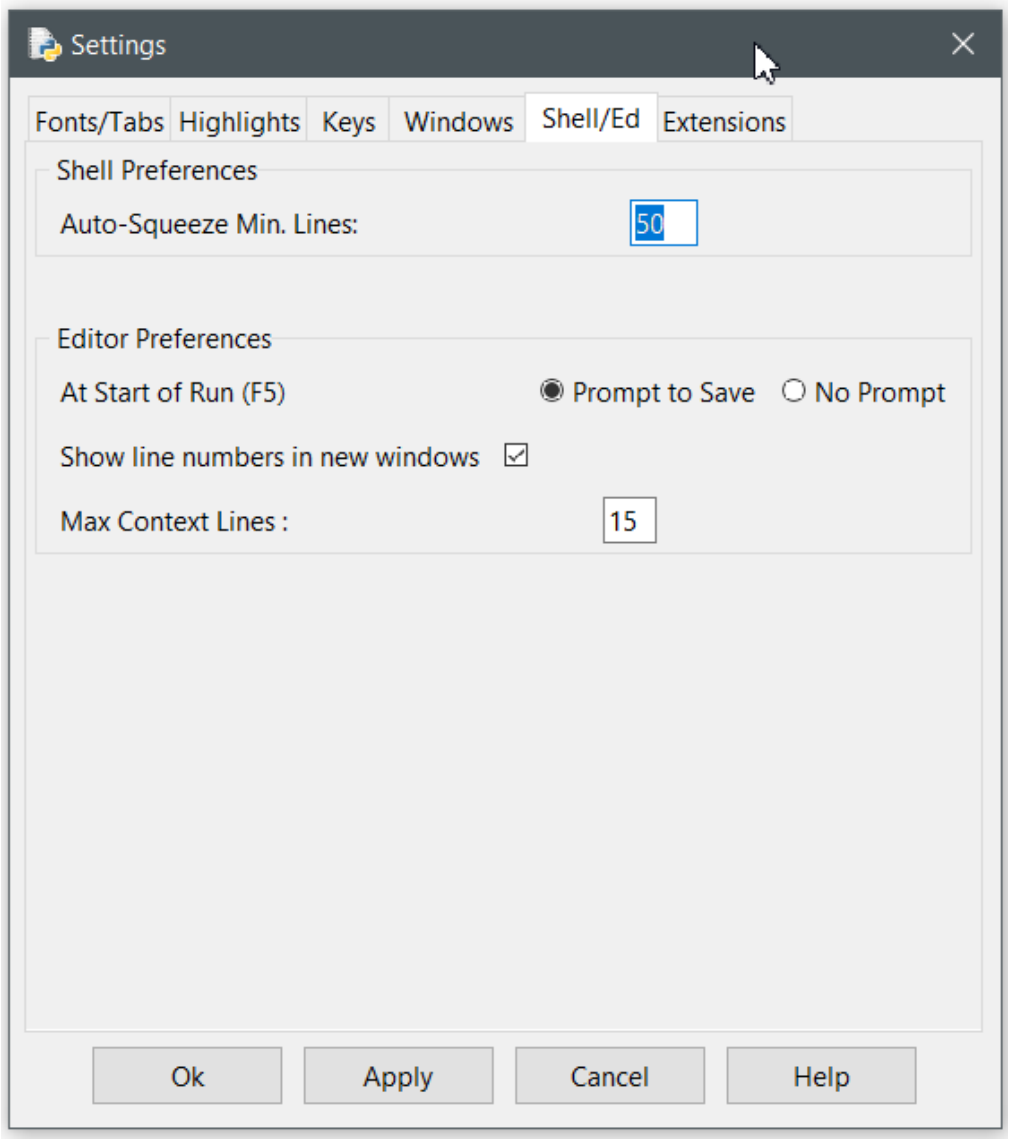

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python
for
windows
Close



By Nishan Saliya


C:\Windows\system32\cmd.exe - python
Microsoft Windows [Version 10.0.19043.1202]
(c) Microsoft Corporation. All rights reserved.
C:\Users\Nishan>python
Python 3.9.7 (tags/v3.9.7:1016ef3, Aug 30 2021, 20:19:38) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
>>>

Python IDLE


Settings
Fonts/Tabs
Highlights
Keys
Windows
Shell/Ed
Extensions
Shell Preferences
Auto-Squeeze Min. Lines:
50
Editor Preferences
At Start of Run (F5)
Prompt to Save
No Prompt
Show line numbers in new windows
Max Context Lines :
15
Ok
Apply
Cancel
Help

Basic of Pseudocodes

- English like syntax
- Write the algorithm in point form with numbers
- Highly compatible with high-level languages
  - C, C++, Java, Python
- But we prefer to use more *code-like* approach

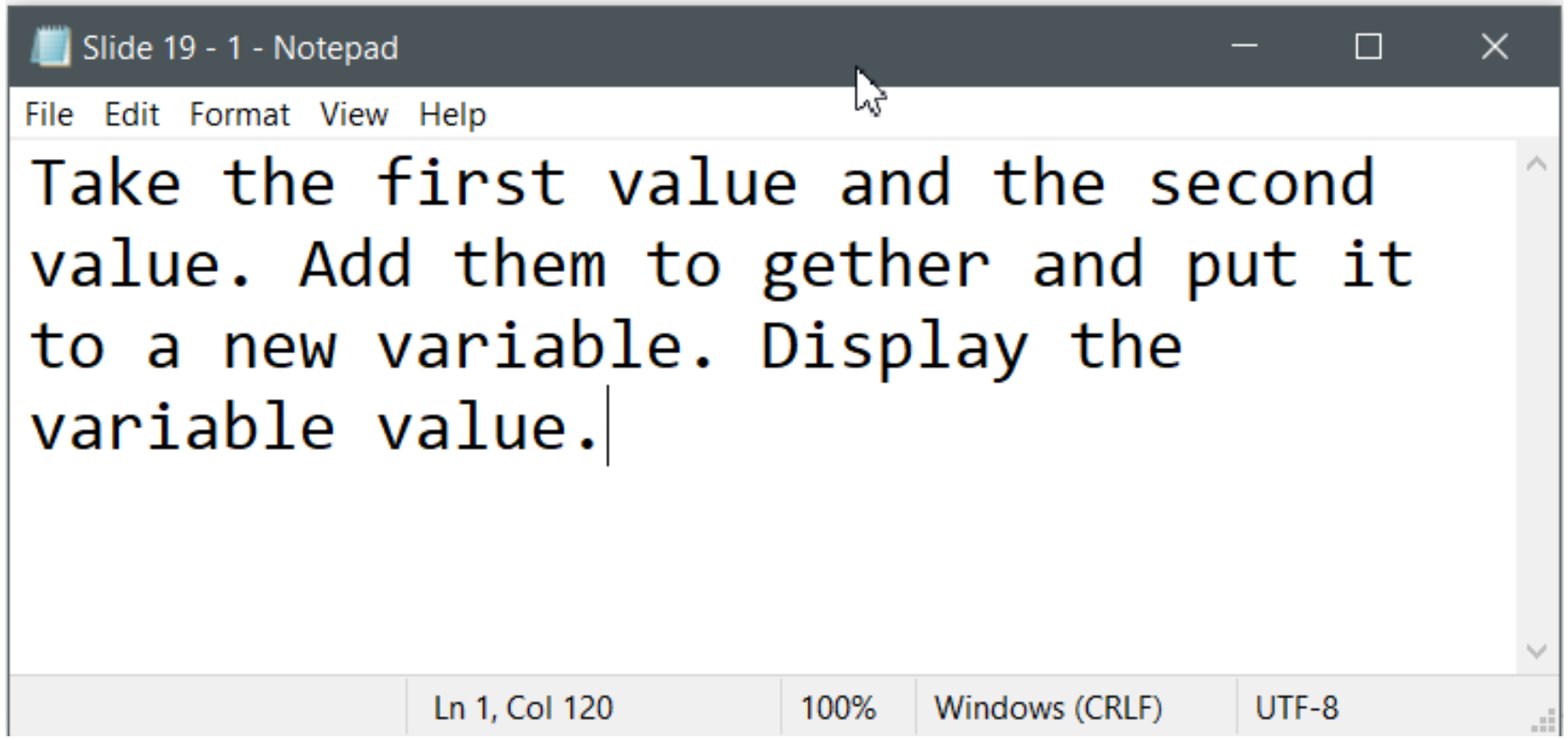
The Problem



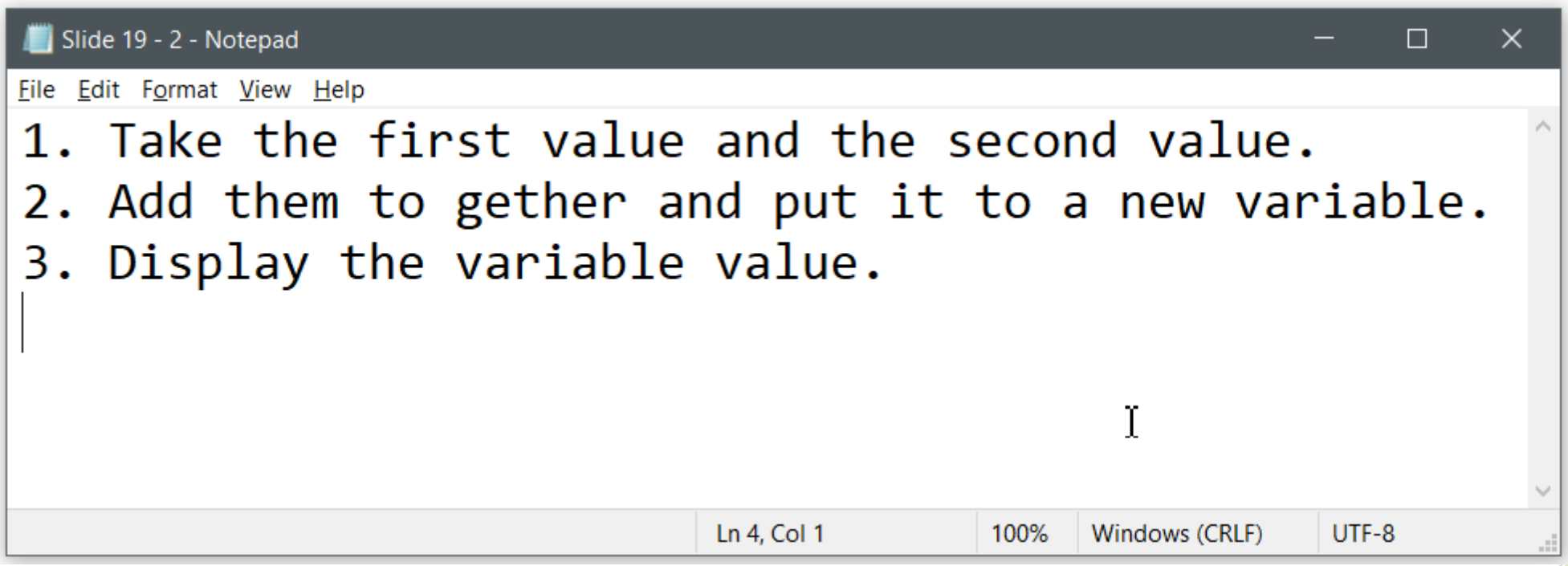
- *Get two numbers from user and add them. Display the answer back to user.*

1. *Write as in a paragraph*
2. *Write as English-like (With steps)*
3. *Write as code-like (With steps)*
4. *Convert to flowchart*
5. *Convert to Python code*
   - *Enhance the code (Python & modify the pseudocode)*

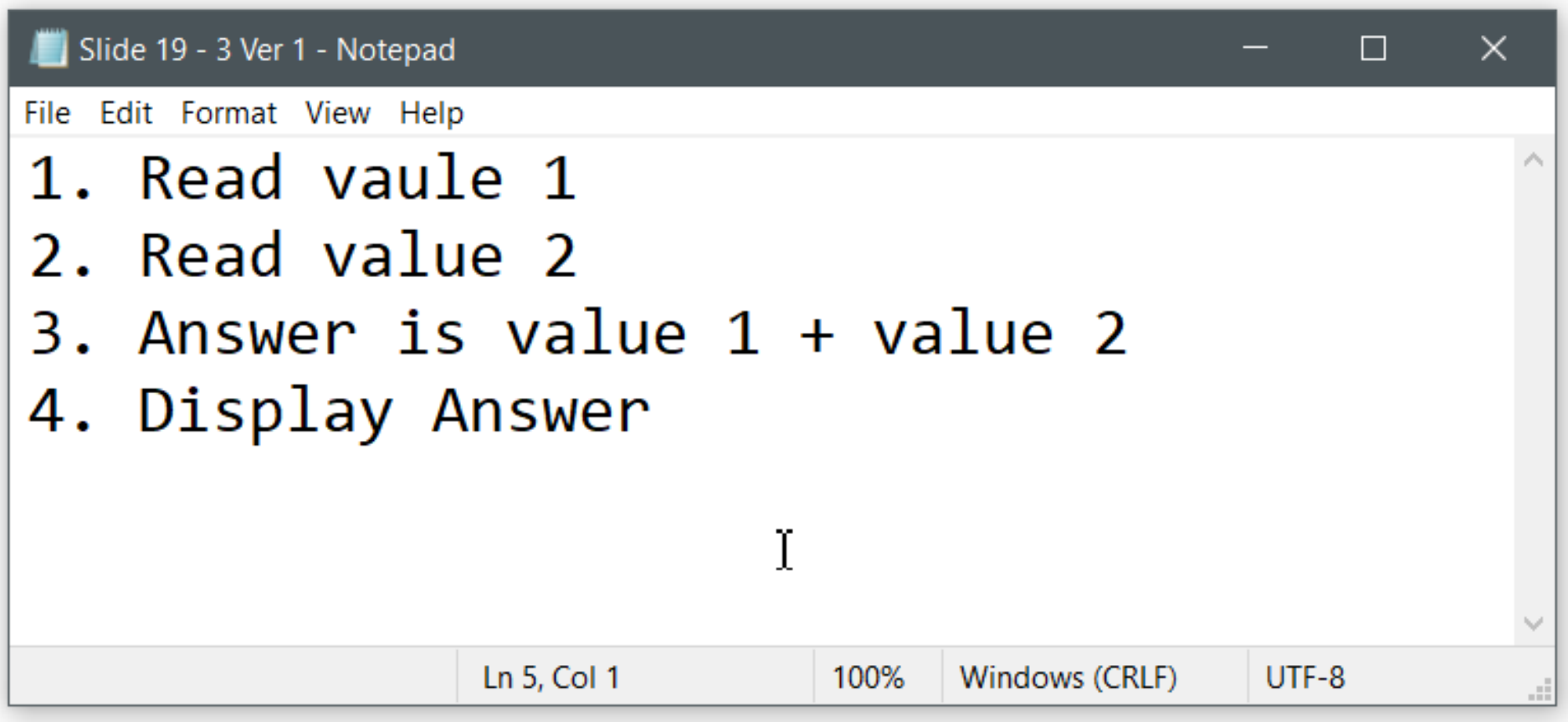
Write as in a paragraph



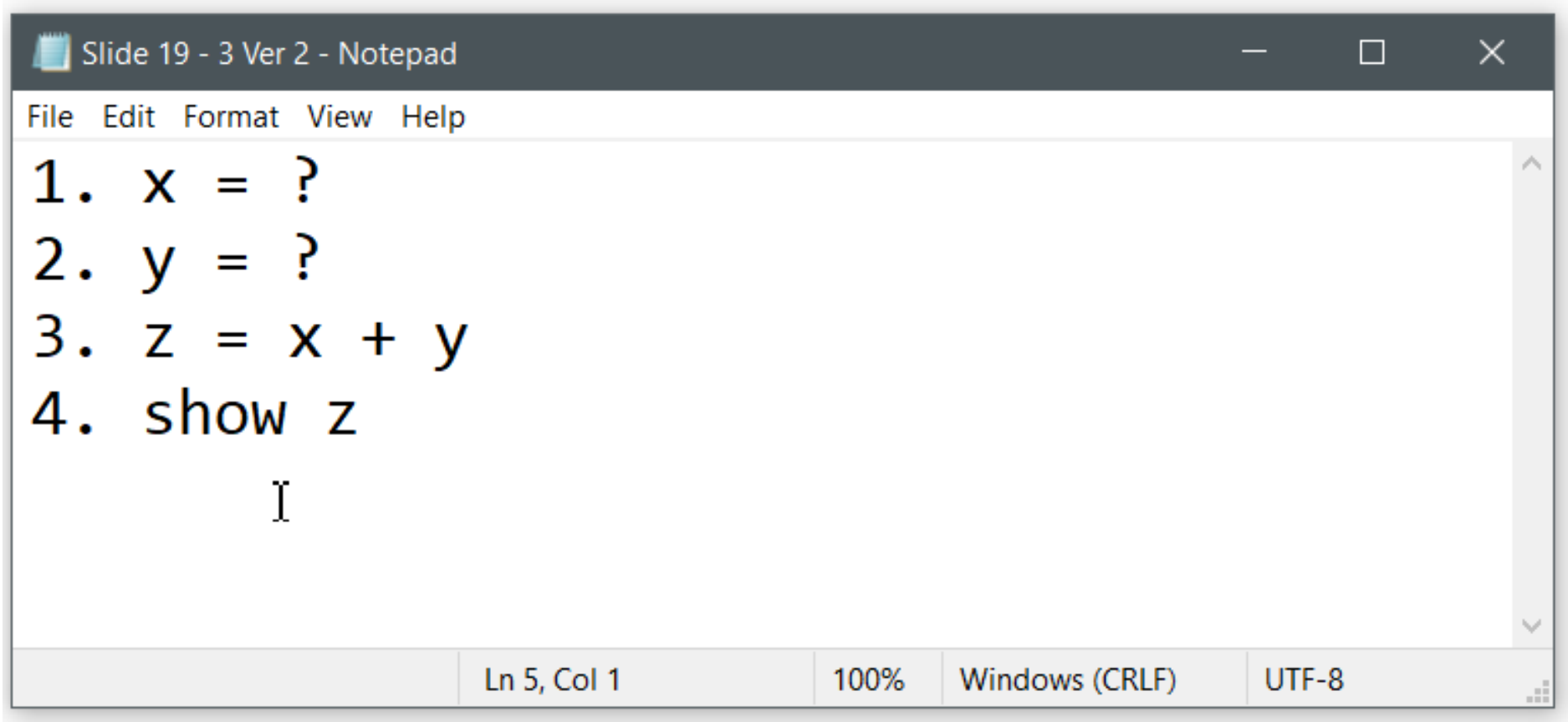
Write as English-like (With steps)



Write as code-like (With steps) Version 1

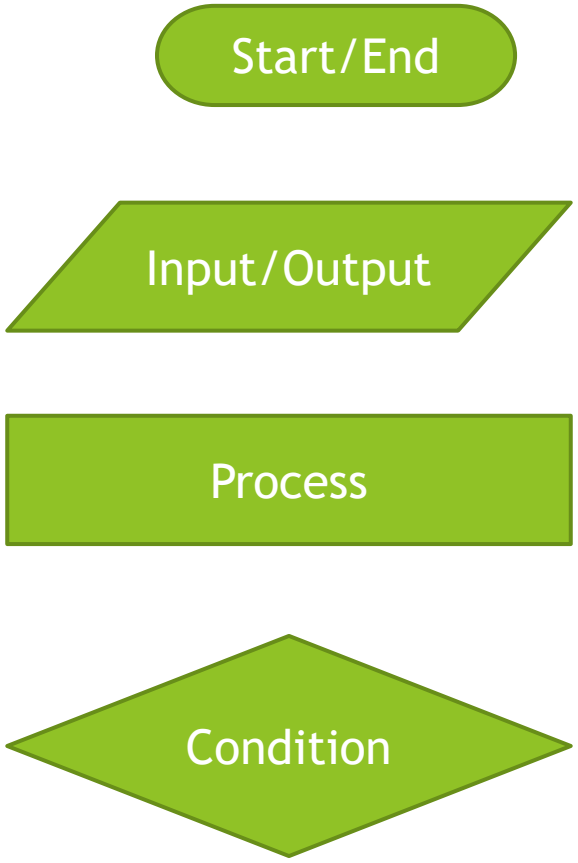


Write as code-like (With steps) Version 2



Basic of Flow Charts

Basic Flowchart Symbols





Flow lines



Connector

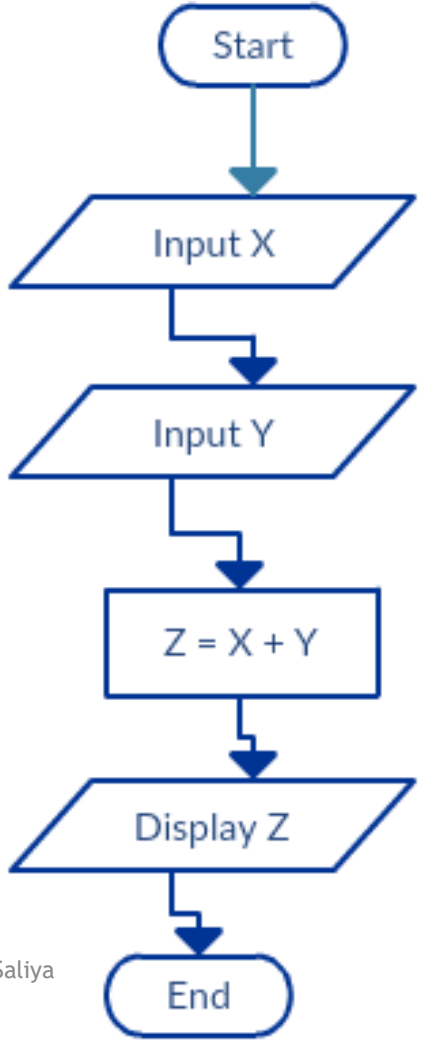


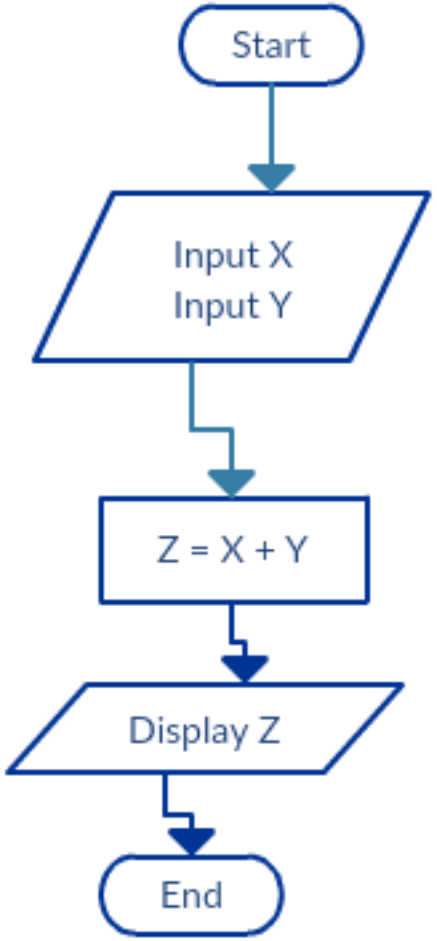
Off page connector

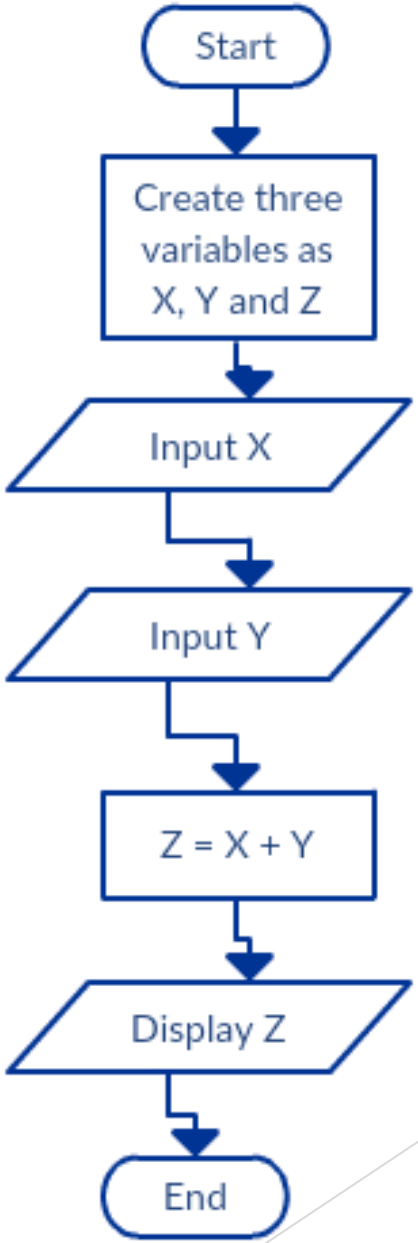
Some Online Tools

- Recommended
  - https://creately.com/
- Alternatives
  - https://www.gliffy.com/
  - https://www.draw.io/ (no account needed)
  - http://diagramo.com/ (no account needed)

Flowchart Versions





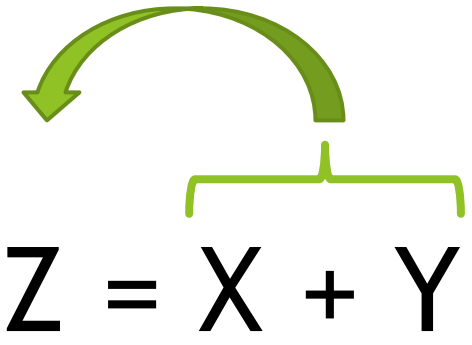


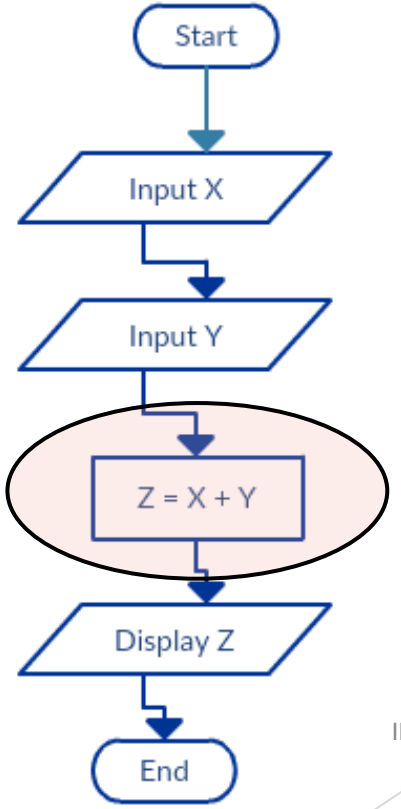
Some Norms for Variables

- Use a proper name for variables
  - Instead of x and you can say Age
- Try not to use spaces for variable names
  - Instead of x and y you can say,
    - Number_1 and Number_2
    - Num1 and Num2
    - NumberOne and NumberTwo
  - Underscore or combined letters may help (even camel case)
- This practice will become handy when you start programming

Equal Sign in Programming

- = means “assign” …. Not exactly the general mathematical EQUAL

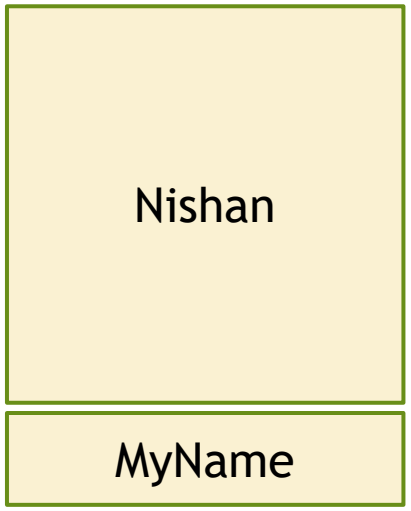




- This may also be used to compare!

Assign Values

MyName = “Nishan”



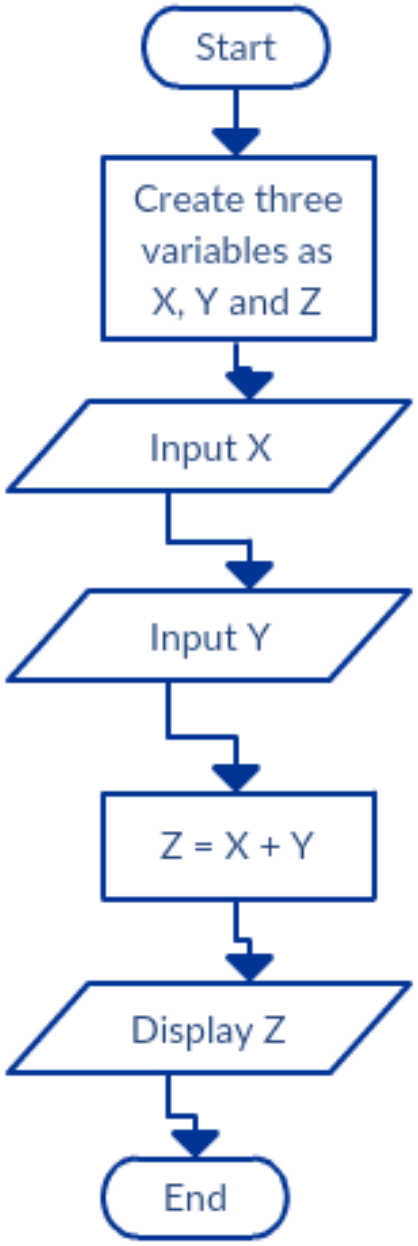
Length = 25



Length = 5
Width = 5
Area = Length x Width



Final Flowchart



Develop the Python Code for Question



- Program 1
  - Very simple. Hardcoded values.
- Program 2
  - Very simple. Dynamic values.
- Program 3
  - Better readable variables
- Program 4
  - Comments
  - Higher maintainability of code
- Program 5
  - Better user experience
  - Runs in console too!

Good Program …

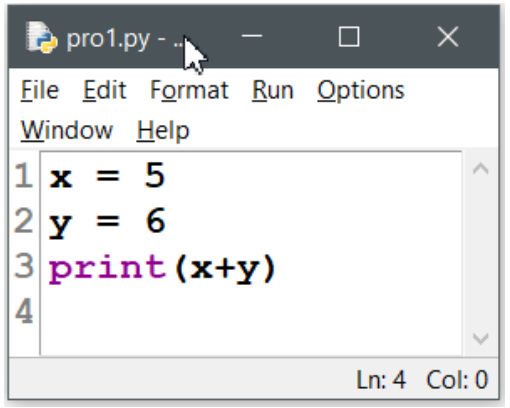
- Properly written algorithm
  - Pseudocode or Flowcharts
  - Maintainable code
  - Better user experience

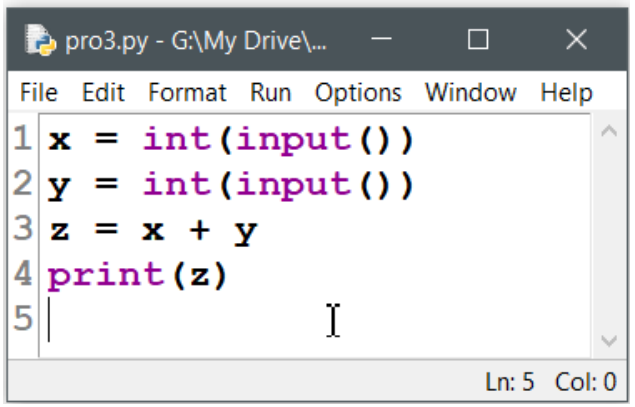
Sample marking sheet

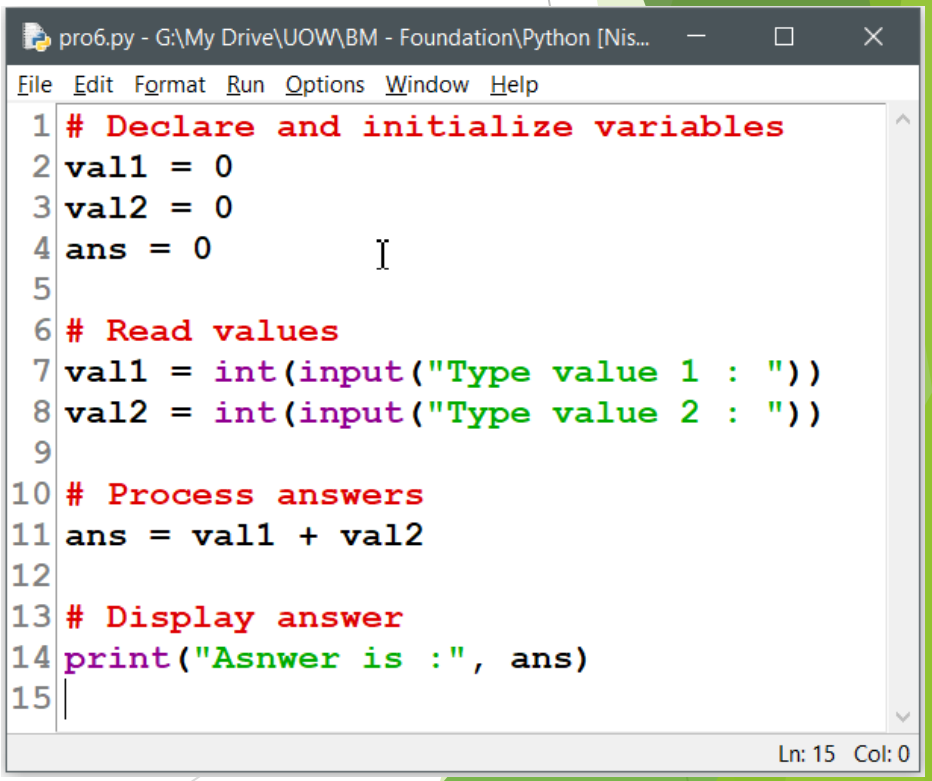
	A	B	C	D
1	**No**	**Description**	**Marks (10 Max)**	**Note**
2	1	Proper input read	10	Dynamic values
3	2	Proper comments	10	No overkilling!
4	3	Proper varaibale names	10	Valid and comprehensible. Must increase readability
5	4	Proper opertaion	10	Works for algorythem
6	5	Proper output	10	Display proper output
7	6	Works in IDLE	10	[[[Generic requirment]]]
8	7	Works in console	10	[[[Specific requirment]]]
9				
10			70	
11				
12		Final marks	100%	

Student 1 Vs Student 2 Marking

Evolution of Your Code







Volume of a cube

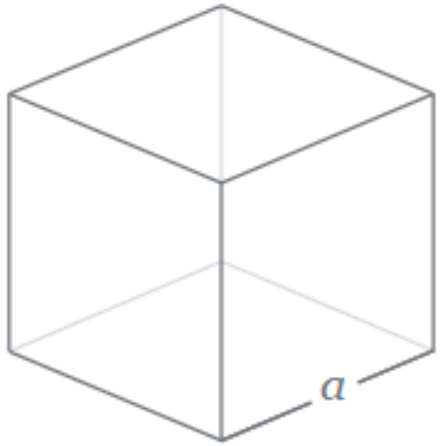
- Write a Python 3.x program to find the volume of a cube
- Take 10 minutes
- Don’t forget good programming practices!

Cube

Solve for volume

$$V = a^3$$

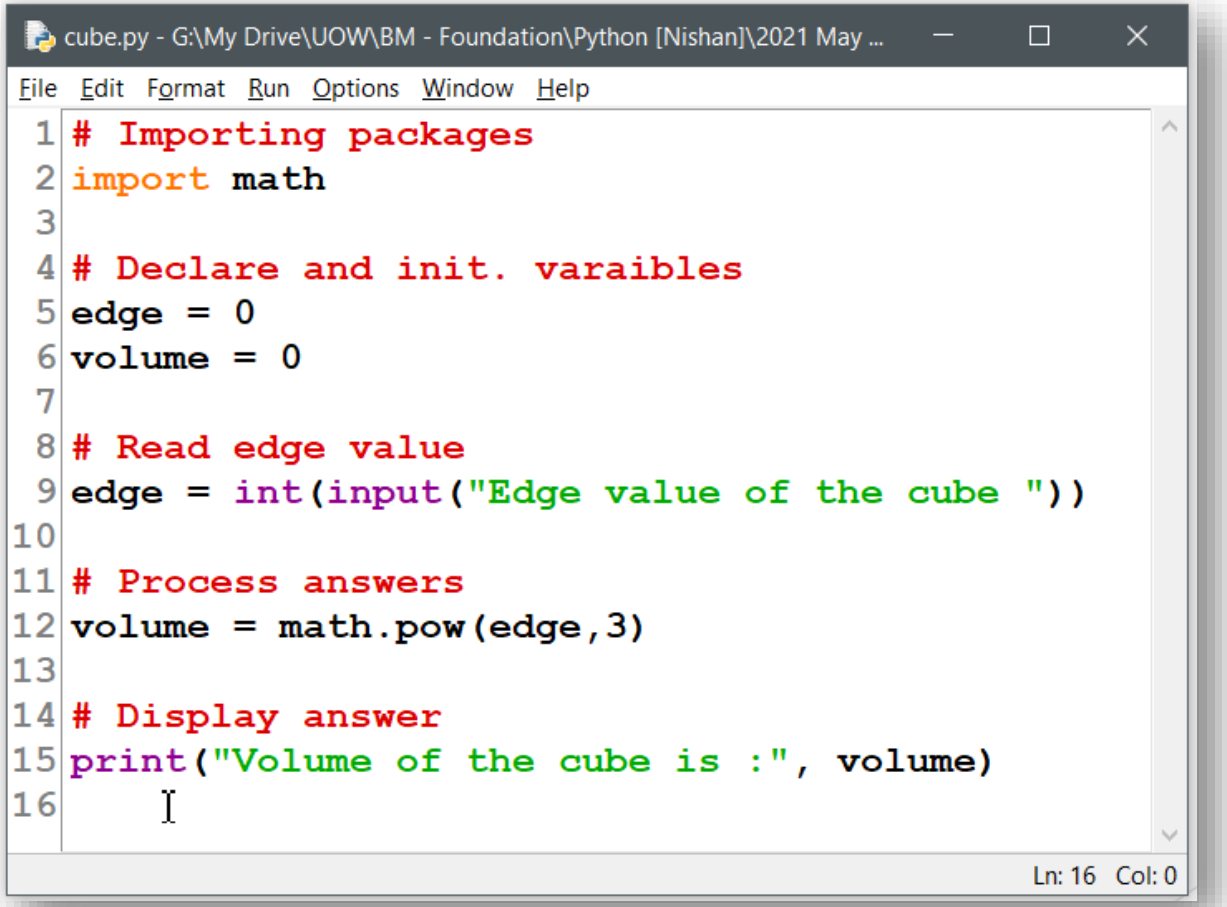
a Edge



Sample Marking Scheme

	A	B	C	D	E
1	**No**	**Description**	**Marks (10 Max)**	**Student Marks**	**Note**
2	1	Import packages	10		Math package [[[Only for this]]]
3	2	Declare & init. Vars	10		Declare and initialize before use
4	3	Proper input read	10		Dynamic values with message
5	4	Input with a message	10		Proper message for users
6	5	Proper comments	10		No overkilling!
7	6	Proper varaibale names	10		Valid and comprehensible. Must increase readability
8	7	Proper opertaion	10		Works for algorythem
9	8	Proper output	10		Display proper output
10	9	Output with a message	10		Proper message for users
11	10	Works in IDLE	10		[[[Generic requirment]]]
12	11	Works in console	10		[[[Specific requirment]]]
13					
14			110	0	
15					
16		Final marks	100%	0%	

Sample Program (Just Another Version)



Any Questions?